Data Scientist & Economic Analyst

Rodrigo França Chaves

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Website · Google Scholar · LinkedIn · GitHub · Brazilian citizen, F1 visa (STEM OPT eligible)

Applied economist and data scientist working on property taxation, housing markets, transit, and utility regulation. First author of a peer-reviewed article in *Utilities Policy* on the health effects of water privatisation in Brazil. I work with large administrative datasets and with the regulation that governs them, combining quasi-experimental design with machine learning, and I am used to presenting a result to the people who have to act on it.

EDUCATION

M.S. in Computational Analysis and Public Policy *Sep 2025 – Jun 2027*
The University of Chicago, Chicago, IL
Specializations: Markets & Regulation; Health Policy
Coursework: Machine Learning for Public Policy, Applied Econometrics, Data Engineering, Program Evaluation, Public Finance, Statistical Computing

Postgraduate Degree in Data Science (Pós-Graduação) *2024*
Insper – Institute of Research and Education, São Paulo, Brazil
Coursework: Statistical Inference, Econometrics, Machine Learning, Cloud Computing, Predictive Modeling

B.A. in International Relations *2022*
Ibmec – Brazilian Institute of Capital Markets, Belo Horizonte, Brazil
Coursework: Political Economy, International Trade, Macroeconomics, Quantitative Methods, Public Policy, Political Science

PUBLICATIONS & WORKING PAPERS

Private Ownership of Water and Wastewater Systems: Assessing Health Impacts
Utilities Policy, peer-reviewed journal article (first author)
Brazil's 2020 sanitation law made it substantially easier for private operators to run water and wastewater systems. This paper asks whether the earlier wave of privatisations justified that decision, comparing propensity-matched municipalities before and after transfer between 1998 and 2021 with Callaway and Sant'Anna staggered difference-in-differences. We find significant reductions in hospital admissions for diarrheal disease among children under five (ATT -11.6 for infants, -15.8 at ages 1 to 5) and no effect on diseases unrelated to sanitation. Results vary across municipalities; the most robust evidence comes from Tocantins, the only state to move to full private ownership. I wrote the paper, assembled the data, and wrote the analysis code; my co-author supervised.

Body-Worn Cameras and Property Crime: Reduced-Form Evidence from São Paulo
Under review at Cities (second author)
Between 2021 and 2022 São Paulo deployed 10,125 body cameras across 64 police battalions. Using confidential Military Police records from 2020 to 2023, we estimate what changed. Battalion boundaries are not publicly released, so I reconstructed them by clustering geocoded incidents into Voronoi polygons, which is what made spatial treatment assignment possible. Confrontations fell (ATT -0.50, significant at 1%), with the decline concentrated in low-income areas (5%, no significant effect in high-income ones), while property crime rose, most robustly for vehicle theft. The pattern is consistent with de-policing rather than a shift toward procedurally correct enforcement.

WORK EXPERIENCE

Data Science and Research Intern, DECDI (formerly DIME) *Jun – Aug 2026*
World Bank, Washington, DC

- Measured how construction in Dakar responded to a new bus rapid transit corridor, using difference-in-differences across 100 m grid cells built from Google Open Buildings and GHSL satellite data. Because the corridor was announced in 2018 and opened in 2024, the design identifies the anticipation and construction-phase response rather than the effect of operation.
- Identified that the project's building-detection threshold had been set at nearly twice the value its source paper recommends for Africa. The setting discarded real structures most heavily in dense and informal areas, a bias that varies by neighborhood and so does not difference out. Shipped the corrected pipeline with 200 tests and an interactive explorer.
- Reconstructed four decades of Mumbai property tax and land-use policy into a single timeline, then tested millions of assessment records for bunching at policy thresholds, the signature of manipulated declared values.
- Quantified how Mumbai's market for additional floor area divides across regulatory instruments (TDR, Premium FSI, Fungible FSI) over two decades of transactions, and which market, regulatory, and political conditions drive that

choice. *(Findings confidential to the World Bank.)*

Graduate Research Assistant *Mar – Jun 2026 · Resuming Sep 2026*

Mansueto Institute for Urban Innovation, UChicago, Chicago, IL

- Built a digitisation pipeline for century-old fire insurance registers, cloth-bound volumes that had never been machine-readable, covering page segmentation, OCR, and structured extraction.
- Recovered thousands of firm addresses from hundreds of pages, matched them to three historical censuses by street, and classified establishments into hundreds of industries using natural language processing on firm names. The result is a record of industrial location in twentieth-century Chicago that did not previously exist.
- Returning in September 2026 to write the method up as a paper documenting the pipeline for other researchers working with archival records.

Research Assistant *Aug 2023 – Jul 2025*

Insper Cities Lab (Arq.Futuro), São Paulo, Brazil

- Two years of quasi-experimental policy evaluation across water utilities, crime, property taxation, and real estate markets in Brazil, using difference-in-differences, regression discontinuity, and spatial methods.
- Constructed the underlying data: home prices, tax assessments, and disease rates matched across municipalities and census tracts, and grouped into new geographic units by spatial clustering where no official boundary suited the question.
- Reconciled administrative records from several government sources, resolving mismatched identifiers and gaps in coverage so the panels aligned across two decades. Without that step the privatisation analysis would not have been identifiable.
- First author on the resulting *Utilities Policy* article and second author on an in-progress working paper on police body cameras.
- Mentored undergraduate researchers on research design, econometrics, and academic writing, and coordinated teams working to publication deadlines.

Teaching Assistant: GIS, Econometrics and R *Aug 2023 – Jul 2025*

Insper, São Paulo, Brazil

- Taught GIS and R to more than 50 students, using São Paulo as the working case: its housing market, its transit network, and the way density and land values decline with distance from the center.
- Advised undergraduate theses through Brazil's required capstone; one placed second in its cohort.
- Ran an intensive field course conducted with municipal officials.

PROJECTS

NYC Property Tax: A Horizontal-Equity Audit `Python · LightGBM · scikit-learn`

Horizontal equity requires that similar properties carry similar tax burdens. This project tests whether New York's assessment roll meets that standard. A citywide benchmark is uninformative when a Manhattan co-op and a Queens two-family fall into the same comparison, so each property is measured against a peer group defined by borough, building class, tax class, decade built, size, and value band, with a five-level fallback so that rare building types still receive a comparison group. On that definition roughly 43% sit more than 15% from their peer-group median; both the peer definition and the threshold are ours, so the figure measures dispersion within comparable groups rather than proven misassessment. Narrowing the value bands from quintiles to ventiles cut within-group price spread from $475k to $48k, trading a small accuracy loss for labels that carry meaning. Built on 138 leakage-free features and LightGBM, delivered as a dashboard for audit triage. I designed the peer-group methodology, built the data pipeline, and trained the models. [GitHub] [Live Dashboard]

Data Centers Next Door: Cloud Infrastructure and Housing Costs `R · Shiny · Spatial Econometrics`

Data center construction around Chicago has accelerated, raising the question of what it does to nearby housing and household costs. I scraped 124 listed facilities, confirmed and geocoded 45, and dated each opening from air permit filings between 2000 and 2024. Those dates are matched against ZIP-level Zillow home values and Census household and utility costs across 660 metropolitan ZIP codes, so a user can examine what moved around each opening. The design is descriptive rather than causal: a facility opening near rising prices is not evidence that it caused them. On a four-person course project I did the scraping, geocoding, and Shiny dashboard. [GitHub] [Live Dashboard]

TECHNICAL SKILLS

Methods: Causal Inference (DiD, Staggered DiD, RDD, IV), Panel & Spatial Econometrics, Remote Sensing

Programming & Data: Python, R, SQL, PostgreSQL, Stata; data pipelines, OCR, geospatial workflows

Tools: QGIS, ArcGIS, Git, LaTeX, Excel, Streamlit, Shiny

Domain: Property tax & valuation, zoning & land use, housing & infrastructure policy, utility regulation

Languages: Portuguese (Native), English (Fluent, TOEFL 112/120), Spanish (Conversational), German (Basic)